

Tailscale AWS Exit Node Manual

Self-Hosted Regional VPN Architecture & CLI Operations Guide

This technical manual provides complete instructions for provisioning, configuring, and on-demand lifecycle management of a private Tailscale exit node hosted in the AWS São Paulo region (`sa-east-1`).

1. Concept & Architecture

An exit node routes all web browsing and public internet traffic through a designated remote device. By spinning up an AWS EC2 instance in Brazil and linking it to your tailnet, services see your origin as South America while Tailscale abstracts routing mechanics using its coordination server—making static public IPs completely unnecessary.

2. Cost Breakdown & Part-Time Optimization

Running this infrastructure on-demand (e.g., 2 hours a day, ~60 hours monthly) reduces operating expenditures to minimal levels. Because Tailscale handles dynamic public addressing via its overlay network, stopping your instance terminates public IP billing instantly.

Component	Type / Specification	24/7 Estimated Cost	Part-Time (2 hrs/day)
Compute Instance	t3.micro Linux (São Paulo)	~\$7.50 – \$9.50 / mo	~\$0.84 / mo
Public IPv4 Address	Dynamic (Active only when running)	~\$3.60 / mo	~\$0.30 / mo
Data Transfer	Outbound Traffic	\$0.00 (First 100 GB Free)	\$0.00 (Under limit)
Total Estimated Cost	Optimized On-Demand Setup	~\$11.00 – \$13.00 / mo	< \$1.20 / mo

Smart TV Compatibility Note: Android TV and Apple TV (tvOS 17+) natively support official Tailscale apps in their stores, allowing direct exit-node selection. Samsung (Tizen) and LG (webOS) lack native clients; use a dedicated streaming box or a router-level client for those models.

3. Manual Configuration Guide

After launching a standard Ubuntu or Amazon Linux `t3.micro` instance in AWS `sa-east-1`, execute the following configuration steps via SSH to establish the exit node capability:

```
# 1. Install Tailscale
curl -fsSL https://tailscale.com/install.sh | sh

# 2. Enable IP Forwarding in Linux kernel
echo 'net.ipv4.ip_forward = 1' | sudo tee -a /etc/sysctl.d/99-tailscale.conf
echo 'net.ipv6.conf.all.forwarding = 1' | sudo tee -a /etc/sysctl.d/99-tailscale.conf
sudo sysctl -p /etc/sysctl.d/99-tailscale.conf

# 3. Bring Tailscale up advertising exit node status
sudo tailscale up --advertise-exit-node
```

Finally, navigate to your Tailscale Admin Console, locate the instance under Machines, open settings, and check "Use as exit node".

4. Automated Command-Line Lifecycle Scripts

To avoid paying for idle hours, use these AWS CLI snippets to spin up or tear down your São Paulo exit node instantly from your local terminal.

Spin Up Script (Start Instance)

```
#!/bin/bash
INSTANCE_ID="i-0123456789abcdef0" # Replace with your EC2 Instance ID
REGION="sa-east-1"

echo "Starting São Paulo exit node ($INSTANCE_ID)..."
aws ec2 start-instances --instance-ids $INSTANCE_ID --region $REGION

echo "Waiting for instance to reach running state..."
aws ec2 wait instance-running --instance-ids $INSTANCE_ID --region $REGION
echo "Exit node is online! Select it in your Tailscale app."
```

Tear Down Script (Stop Instance)

```
#!/bin/bash
INSTANCE_ID="i-0123456789abcdef0" # Replace with your EC2 Instance ID
REGION="sa-east-1"

echo "Stopping São Paulo exit node to halt billing..."
aws ec2 stop-instances --instance-ids $INSTANCE_ID --region $REGION

echo "Waiting for instance to stop..."
aws ec2 wait instance-stopped --instance-ids $INSTANCE_ID --region $REGION
echo "Exit node shut down successfully. Compute and IP fees stopped."
```